

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

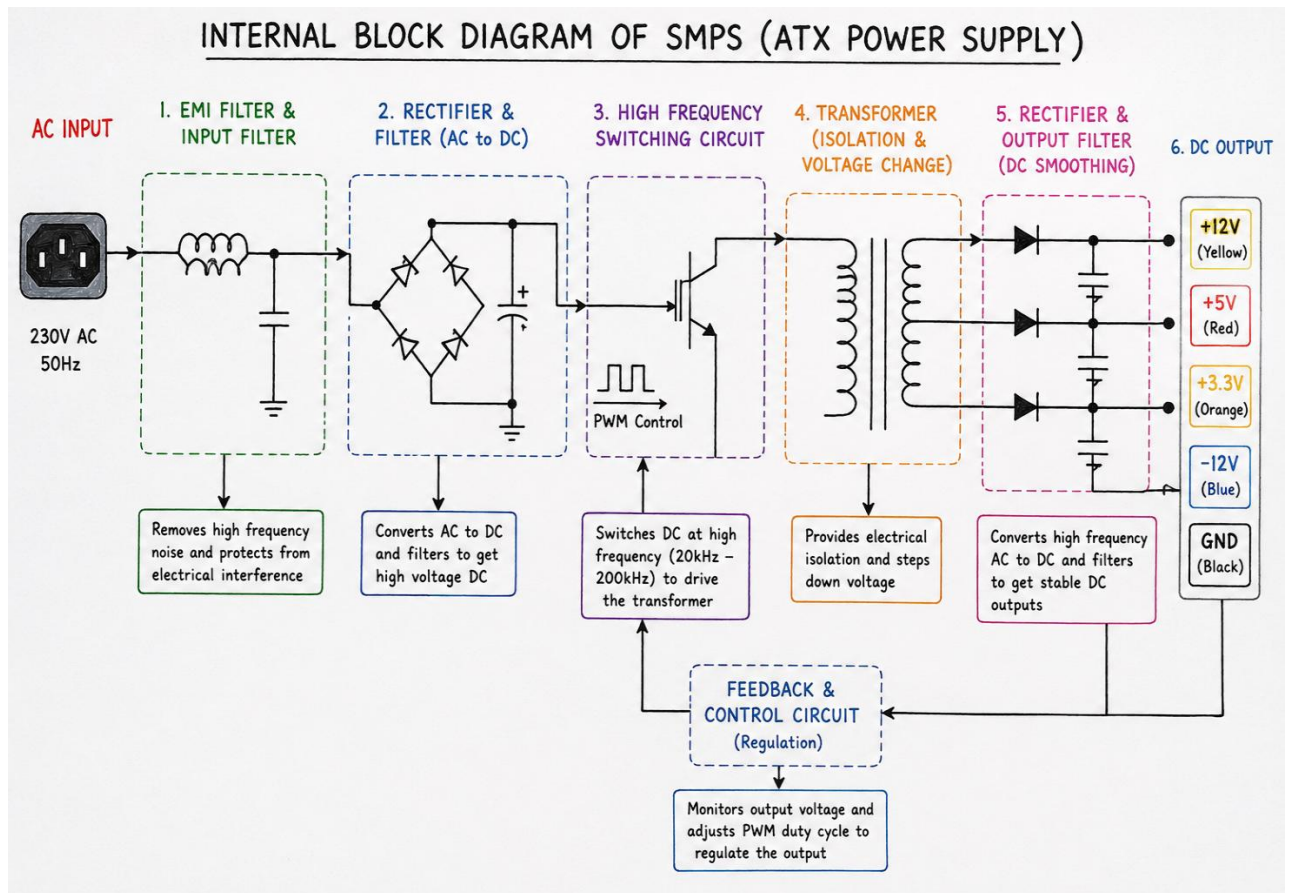
Devices Using SMPS (Switched Mode Power Supply)

Device	Why it Needs SMPS
Desktop Computer	It requires efficient and regulated DC power for its internal components.
LED Television	It converts AC power into stable DC power for smooth operation.
Laptop Charger	It provides efficient voltage conversion and consumes less power.

Devices Using UPS (Uninterruptible Power Supply)

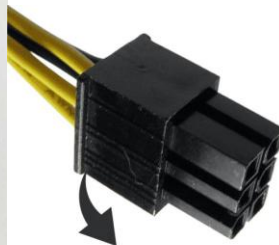
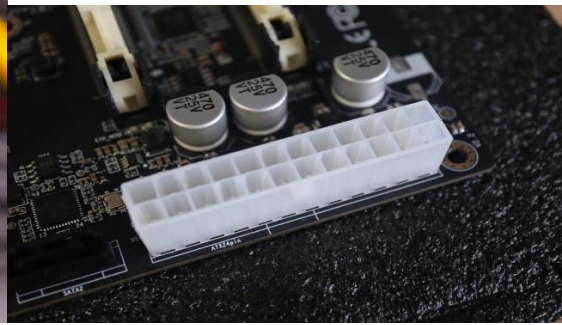
Device	Why it Needs UPS
Desktop Computer	It provides backup power during a power failure to prevent data loss.
Wi-Fi Router	It keeps the internet connection active during short power outages.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Connector Name	Devices It Powers	Safety Tip
ATX 24-pin Motherboard Connector	Supplies power to the motherboard and its components.	Turn off the computer and unplug the power supply before connecting or disconnecting it.
SATA Power Connector	Powers SATA hard drives, SSDs, and optical drives.	Hold the connector by its plastic housing and do not force it into the port.
Molex Connector (4-pin)	Powers older hard drives, CD/DVD drives, and case fans.	Pull the connector straight out to avoid bending the pins.
PCIe 6/8-pin Connector	Provides power to graphics cards (GPUs).	Ensure the connector is fully seated and the PC is powered off before handling it.



6-pin connector



8-pin connector

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Setting Up a New Gaming PC: Connecting the SMPS

Step 1: Connect the Motherboard

- Use the **24-pin ATX power connector** from the SMPS.
- Plug it into the **24-pin socket on the motherboard**.
- This supplies power to the motherboard.

Step 2: Connect the CPU Power

- Use the **8-pin (4+4) CPU power connector**.
- Connect it to the **CPU power socket** near the processor.
- This provides power to the CPU.

Step 3: Connect the Graphics Card

- Use the **PCIe 6-pin or 8-pin power connector**.
- Plug it into the graphics card (GPU).
- This provides additional power required for gaming and high-performance graphics.

Step 4: Connect Storage Devices

- Use the **SATA power connector**.
- Connect it to the **SSD, HDD, or DVD drive**.
- This supplies power to storage devices.

Step 5: Connect Other Devices (Optional)

- Use the **Molex connector** for older hard drives, optical drives, or case fans if required.

Step 6: Check All Connections

- Ensure all connectors are firmly attached.
- Verify that cables are not loose.
- Connect the power cord to the SMPS and turn on the PC.